

features variable $\vee \beta$ Target variable

feature Vector (x) $\rightarrow$ The collection of independent variables mapped as $n \times d$

Target vector (y) $\rightarrow$ The isolated dependent variable for prediction goals

$$\boxed{y = f(x) + c}$$

Data Matters

GIGO $\rightarrow$ Garbage In - Garbage Out

$\Rightarrow$ Bad Data $\rightarrow$ Low prediction

Good Data $\rightarrow$ Complex algorithms

* Mean

$$\boxed{\mu = \frac{1}{n} \sum x_i}$$

} Central Tendency

* Median $\rightarrow$ middle value

Dispersion:

$$\boxed{\text{Variance} = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2 \text{ (me)}}$$

$$\boxed{\text{Standard deviation} = \sqrt{\text{variance}}}$$

* Concept of Distributions

⇒ Normal Distribution → Empirical rule.

68 - 95 - 1.

Skewness

↳ Positive skew → Right tail
↳ Negative skew → Left tail

Kurtosis

↳ High → More outliers
↳ Low → Lower outliers

Correlation

$$r = \frac{\text{Cov}(x, y)}{s_x s_y}$$

Covariance

$$\text{Cov}(x, y) = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{n-1}$$

Multicollinearity Danger

↳ High linear correlation between features
optimization

Missing values

`df.isna().sum()` → detect

Duplicates

Overfitting → Cause overweighing

Outliers

Diminutionality Reduction

PCA → Principal Component analysis

Generalization Error formula

$$\text{Total Error} = \text{Bias}^2 + \text{Variance} + \text{noise}$$

Managing variance - Cross validation

Split the data into k subset

k -Fold Cross validation

↳ Train these k -items

↳ Average the Results